Python Functions

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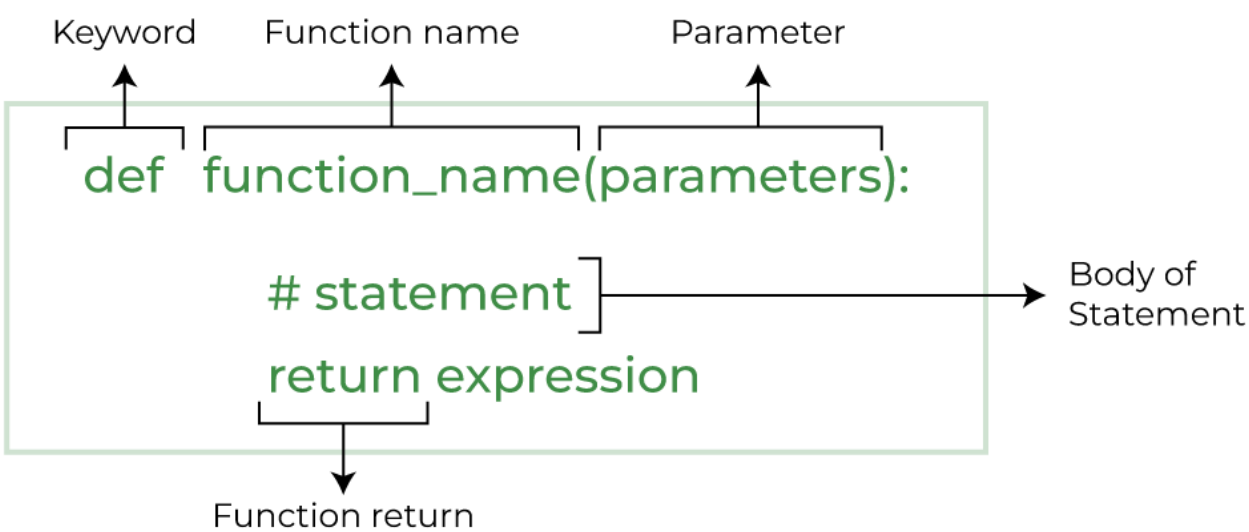
Python Functions is a block of statements that return the specific task. The idea is to put some commonly or repeatedly done tasks together and make a function so that instead of writing the same code again and again for different inputs, we can do the function calls to reuse code contained in it over and over again.
Some **Benefits of Using Functions**

- Increase Code Readability
- Increase Code Reusability

Python Function Declaration

The syntax to declare a function is:



Syntax of Python Function Declaration

Types of Functions in Python

There are mainly two types of functions in Python.

- **Built-in library function:** These are Standard functions in Python that are available to use.
- **User-defined function:** We can create our own functions based on our requirements.

Creating a Function in Python

We can define a function in Python, using the **def** keyword. We can add any type of functionalities and properties to it as we require.

- Python3

```
# A simple Python function

def fun():
  print("Welcome to GFG")
```

Calling a Python Function

After creating a function in Python we can call it by using the name of the function followed by parenthesis containing parameters of that particular function.

- Python3

```
# A simple Python function
def fun():
    print("Welcome to GFG")


# Driver code to call a function
fun()
```

Output:

```
Welcome to GFG
```

Python Function with Parameters

If you have experience in C/C++ or Java then you must be thinking about the *return type* of the function and *data type* of arguments. That is possible in Python as well (specifically for Python 3.5 and above).

Defining and calling a function with parameters

```
def function_name(parameter: data_type) -> return_type:
```

```
    """Docstring"""
    # body of the function
    return expression
```

The following example uses [arguments and parameters] that you will learn later in this article so you can come back to it again if not understood.

- Python3

```python
def add(num1: int, num2: int) -> int:
    """Add two numbers"""
    num3 = num1 + num2

    return num3

# Driver code
num1, num2 = 5, 15
ans = add(num1, num2)
print(f"The addition of {num1} and {num2} results {ans}.")
```

Output:

```
The addition of 5 and 15 results 20.
```

Note: *The following examples are defined using syntax 1, try to convert them in syntax 2 for practice.*

- Python3

```python
# some more functions
def is_prime(n):
```

```
    if n in [2, 3]:

        return True

    if (n == 1) or (n % 2 == 0):

        return False

    r = 3

    while r * r <= n:

        if n % r == 0:

            return False

        r += 2

    return True

print(is_prime(78), is_prime(79))
```

Output:

```
False True
```

Python Function Arguments

Arguments are the values passed inside the parenthesis of the function. A function can have any number of arguments separated by a comma.

In this example, we will create a simple function in Python to check whether the number passed as an argument to the function is even or odd.

- Python3

```
# A simple Python function to check

# whether x is even or odd

def evenOdd(x):
```

```
    if (x % 2 == 0):

        print("even")

    else:

        print("odd")




# Driver code to call the function

evenOdd(2)

evenOdd(3)
```

Output:

```
even
odd
```

Types of Python Function Arguments

Python supports various types of arguments that can be passed at the time of the function call. In Python, we have the following 4 types of function arguments.

- **Default argument**
- **Keyword arguments (named arguments)**
- **Positional arguments**
- **Arbitrary arguments** (variable-length arguments *args and **kwargs)

Let's discuss each type in detail.

Default Arguments

A default argument is a parameter that assumes a default value if a value is not provided in the function call for that argument. The following example illustrates Default arguments.

- Python3

```
# Python program to demonstrate

# default arguments
```

```
def myFun(x, y=50):

    print("x: ", x)

    print("y: ", y)




# Driver code (We call myFun() with only

# argument)

myFun(10)
```

Output:

```
x:  10
y:  50
```

Like C++ default arguments, any number of arguments in a function can have a default value. But once we have a default argument, all the arguments to its right must also have default values.

Keyword Arguments

The idea is to allow the caller to specify the argument name with values so that the caller does not need to remember the order of parameters.

- Python3

```
# Python program to demonstrate Keyword Arguments

def student(firstname, lastname):

    print(firstname, lastname)




# Keyword arguments
```

```
student(firstname='Geeks', lastname='Practice')

student(lastname='Practice', firstname='Geeks')
```

Output:

```
Geeks Practice
Geeks Practice
```

Positional Arguments

We used the [Position argument] during the function call so that the first argument (or value) is assigned to name and the second argument (or value) is assigned to age. By changing the position, or if you forget the order of the positions, the values can be used in the wrong places, as shown in the Case-2 example below, where 27 is assigned to the name and Suraj is assigned to the age.

- Python3

```
def nameAge(name, age):

    print("Hi, I am", name)

    print("My age is ", age)




# You will get correct output because

# argument is given in order

print("Case-1:")

nameAge("Suraj", 27)

# You will get incorrect output because

# argument is not in order

print("\nCase-2:")
```

```
nameAge(27, "Suraj")
```

Output:

```
Case-1:
Hi, I am Suraj
My age is  27
Case-2:
Hi, I am 27
My age is  Suraj
```

Arbitrary Keyword Arguments

In Python Arbitrary Keyword Arguments, *args, and **kwargs can pass a variable number of arguments to a function using special symbols. There are two special symbols:

- *args in Python (Non-Keyword Arguments)
- **kwargs in Python (Keyword Arguments)

Example 1: Variable length non-keywords argument

- Python3

```
# Python program to illustrate

# *args for variable number of arguments

def myFun(*argv):

    for arg in argv:

        print(arg)




myFun('Hello', 'Welcome', 'to', 'GeeksforGeeks')
```

Output:

```
Hello
Welcome
```

```
to
GeeksforGeeks
```

Example 2: Variable length keyword arguments

- Python3

```
# Python program to illustrate
# *kwargs for variable number of keyword arguments


def myFun(**kwargs):
    for key, value in kwargs.items():
        print("%s == %s" % (key, value))


# Driver code
myFun(first='Geeks', mid='for', last='Geeks')
```

Output:

```
first == Geeks
mid == for
last == Geeks
```

Docstring

The first string after the function is called the Document string or [Docstring]() in short. This is used to describe the functionality of the function. The use of docstring in functions is optional but it is considered a good practice.

The below syntax can be used to print out the docstring of a function:

Syntax: `print(function_name.__doc__)`

Example: Adding Docstring to the function

- Python3

```
# A simple Python function to check
# whether x is even or odd


def evenOdd(x):
    """Function to check if the number is even or odd"""

    if (x % 2 == 0):
        print("even")
    else:
        print("odd")


# Driver code to call the function
print(evenOdd.__doc__)
```

Output:

```
Function to check if the number is even or odd
```

Python Function within Functions

A function that is defined inside another function is known as the **inner function** or **nested function**. Nested functions can access variables of the enclosing scope. Inner functions are used so that they can be protected from everything happening outside the function.

- Python3

```
# Python program to
# demonstrate accessing of
# variables of nested functions


def f1():
    s = 'I love GeeksforGeeks'

    def f2():
        print(s)

    f2()

# Driver's code
f1()
```

Output:

```
I love GeeksforGeeks
```

Anonymous Functions in Python

In Python, an [anonymous function](#) means that a function is without a name. As we already know the def keyword is used to define the normal functions and the lambda keyword is used to create anonymous functions.

- Python3

```
# Python code to illustrate the cube of a number

# using lambda function

def cube(x): return x*x*x



cube_v2 = lambda x : x*x*x



print(cube(7))

print(cube_v2(7))
```

Output:

```
343
343
```

Recursive Functions in Python

Recursion in Python refers to when a function calls itself. There are many instances when you have to build a recursive function to solve **Mathematical and Recursive Problems.**

Using a recursive function should be done with caution, as a recursive function can become like a non-terminating loop. It is better to check your exit statement while creating a recursive function.

- Python3

```
def factorial(n):

    if n == 0:

        return 1
```

```
    else:

        return n * factorial(n - 1)



print(factorial(4))
```

Output

```
24
```

Here we have created a recursive function to calculate the factorial of the number. You can see the end statement for this function is when n is equal to 0.

Return Statement in Python Function

The function return statement is used to exit from a function and go back to the function caller and return the specified value or data item to the caller. **The syntax for the return statement is:**

```
return [expression_list]
```

The return statement can consist of a variable, an expression, or a constant which is returned at the end of the function execution. If none of the above is present with the return statement a None object is returned.

Example: Python Function Return Statement

- Python3

```
def square_value(num):

    """This function returns the square

    value of the entered number"""

    return num**2




print(square_value(2))
```

```
print(square_value(-4))
```

Output:

```
4
16
```

Pass by Reference and Pass by Value

One important thing to note is, in Python every variable name is a reference. When we pass a variable to a function, a new reference to the object is created. Parameter passing in Python is the same as reference passing in Java.

- Python3

```
# Here x is a new reference to same list lst

def myFun(x):

    x[0] = 20




# Driver Code (Note that lst is modified

# after function call.

lst = [10, 11, 12, 13, 14, 15]

myFun(lst)

print(lst)
```

Output:

```
[20, 11, 12, 13, 14, 15]
```

When we pass a reference and change the received reference to something else, the connection between the passed and received parameters is broken. For example, consider the below program as follows:

- Python3

```
def myFun(x):

    # After below line link of x with previous
    # object gets broken. A new object is assigned
    # to x.
    x = [20, 30, 40]


# Driver Code (Note that lst is not modified
# after function call.
lst = [10, 11, 12, 13, 14, 15]
myFun(lst)
print(lst)
```

Output:

```
[10, 11, 12, 13, 14, 15]
```

Another example demonstrates that the reference link is broken if we assign a new value (inside the function).

- Python3

```
def myFun(x):
```

```
    # After below line link of x with previous

    # object gets broken. A new object is assigned

    # to x.

    x = 20




# Driver Code (Note that x is not modified

# after function call.

x = 10

myFun(x)

print(x)
```

Output:

```
10
```

Exercise: Try to guess the output of the following code.

- Python3

```
def swap(x, y):

    temp = x

    x = y

    y = temp
```

```
# Driver code

x = 2

y = 3

swap(x, y)

print(x)

print(y)
```

Output:

```
2
3
```

Python None Keyword

- None is used to define a **null value or Null object in Python**. It is not the same as an empty string, a False, or a zero. **It is a data type of the class NoneType object.**

None in Python

Python None is the function returns when there are no return statements.

- Python3

```
def check_return():

    pass

print(check_return())
```

Output:

```
None
```

Null Vs None in Python

None – None is an instance of the NoneType object type. And it is a particular variable that has no objective value. While new NoneType objects cannot be generated, None can be assigned to any variable.
Null – There is no null in Python, we can use None instead of using null values.
Note: *Python null is refer to None which is instance of the NoneType object type*

Example:
Here we will both Null and None and we get the related output for these two statements. Where one of the outputs is Null is not defined.

- Python3

```
print(type(None))

print(type(Null))
```

Output:

```
<class 'NoneType'>
-----------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Input In [9], in <cell line: 2>()
      1 print(type(None))
----> 2 print(type(Null))
NameError: name 'Null' is not defined
```

Referring to the null object in Python

Assigning a value of None to a variable is one way to reset it to its original, empty state.

Example 1:
We will check the type of None

- Python3

```
print(type(None))
```

Output:

```
<class 'NoneType'>
```

Example 2:
Declaring a variable as None.

- Python3

```
var = None

# checking it's value
if var is None:
    print("var has a value of None")
else:
    print("var has a value")
```

Output:

```
var has a value of None
```

Note: If a function does not return anything, it returns None in Python.